27 Bacteria and Archaea

KEY CONCEPTS

Study Tip

Make a table: As you read the chapter, identify examples of challenging environments in which prokaryotes thrive. List the structural, metabolic, or other features of prokaryotes that contribute to their success in each of these environments.

Environmental challenge	Prokaryotic feature enabling success
Lack of water	Capsule or slime layer
Exposure to antibiotics	

Go to Mastering Biology

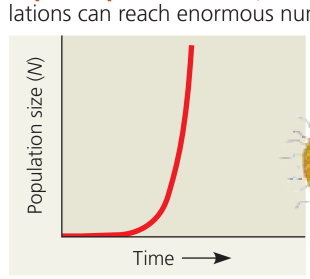
For Students (in eText and Study Area)
- Get Ready for Chapter 27
- Figure 27.13 Walkthrough: Conjugation and Recombination in *E. coli*
- Animation: Structure and Reproduction of Bacteria

For Instructors to Assign (in Item Library)
- Scientific Skills Exercise: Calculating and Interpreting Means and Standard Errors
- Tutorial: Genetic Transfer in Bacteria

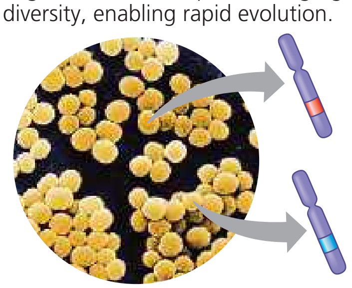
Figure 27.1 The pink color of this lake in Spain is a sign that its waters are much saltier than seawater. Its color is caused not by minerals or other nonliving sources, but by trillions of prokaryotes that thrive in salinities that kill other cells. Other prokaryotes live in environments that are far too cold or hot or acidic for most other organisms.

What characteristics enable prokaryotes to reach huge population sizes and thrive in diverse environments?

Due to the organisms' **small size** and **rapid reproduction**, prokaryotic populations can reach enormous numbers.



Because prokaryotic populations are so large, **mutations** produce high genetic diversity, enabling rapid evolution.



Their **diverse adaptations** enable prokaryotes to live in a wide range of environments.

Rapid evolution in prokaryotic populations results in diverse metabolic and structural adaptations.